

SMART MANAGER ERP

Senior Developer Architecture Diagram Set

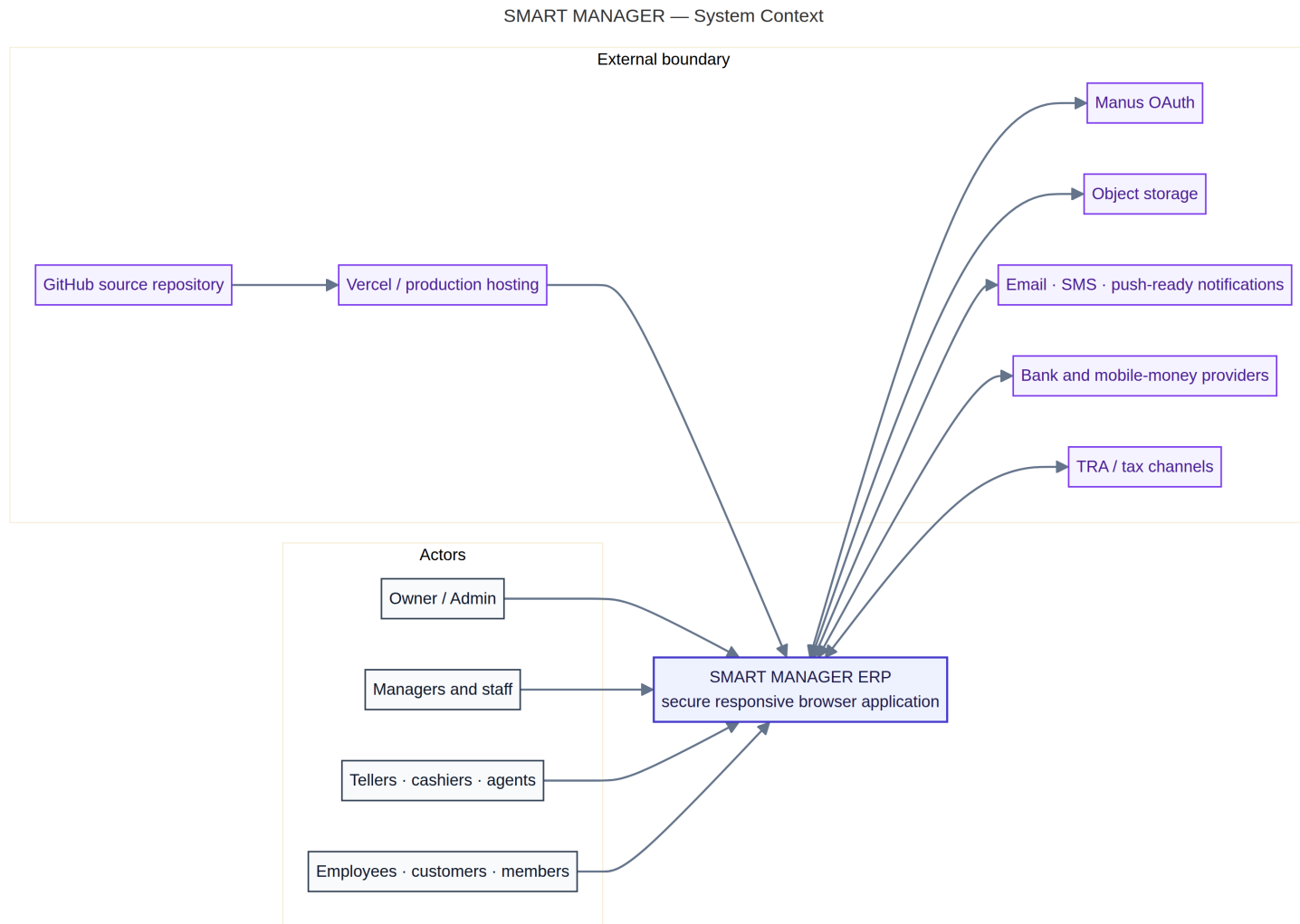
Eight focused views for understanding system boundaries, module composition, financial controls, security, integrations, deployment, and end-to-end workflows.

View	Architectural question answered
1. System context	What is inside the product and what remains external?
2. Application layers	How does a request move from UI to persistent data?
3. Module ecosystem	How are the ERP modules grouped and connected?
4. Shared financial and data plane	How do operational events become controlled financial records?
5. Security, tenant isolation, and governance	Where are identity, isolation, authorization, and audit enforced?
6. Integrations, automation, and scheduled execution	How are providers and recurring processes safely coordinated?
7. Deployment and runtime topology	How does code become a verified production runtime?
8. Critical end-to-end workflows	How do critical journeys cross shared controls?

Design intent: the diagrams are deterministic Mermaid architecture sources rendered as individual high-resolution PNGs. The set favors clear boundaries and reviewable relationships over one overloaded canvas.

1. System context

SMART MANAGER ERP · focused architecture view

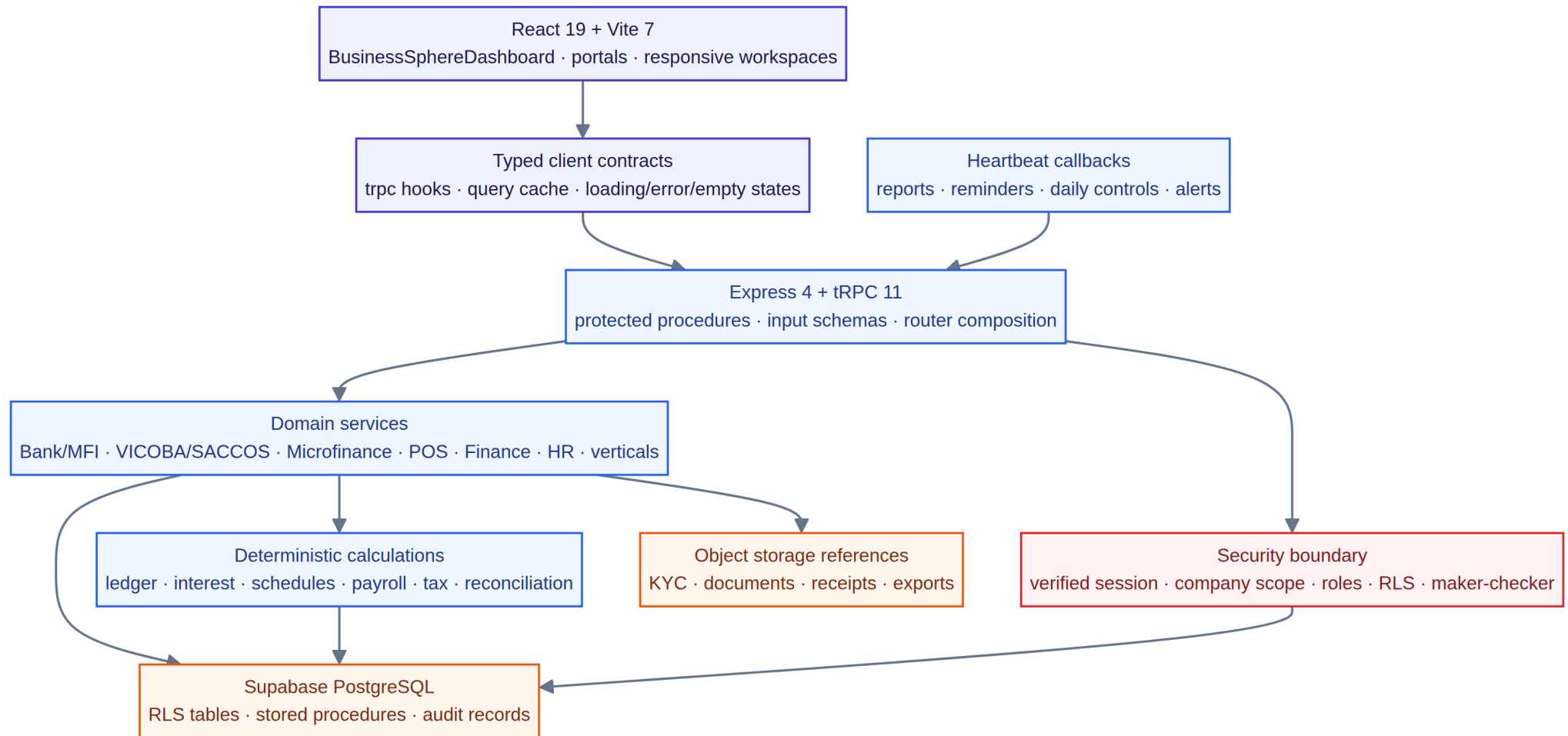


Senior developer interpretation. The context view defines the product boundary. Human users work through the ERP, while identity, storage, notifications, financial providers, tax channels, source control, and hosting remain explicit external dependencies. This prevents the application from being mistaken for the provider systems it orchestrates.

2. Application layers

SMART MANAGER ERP · focused architecture view

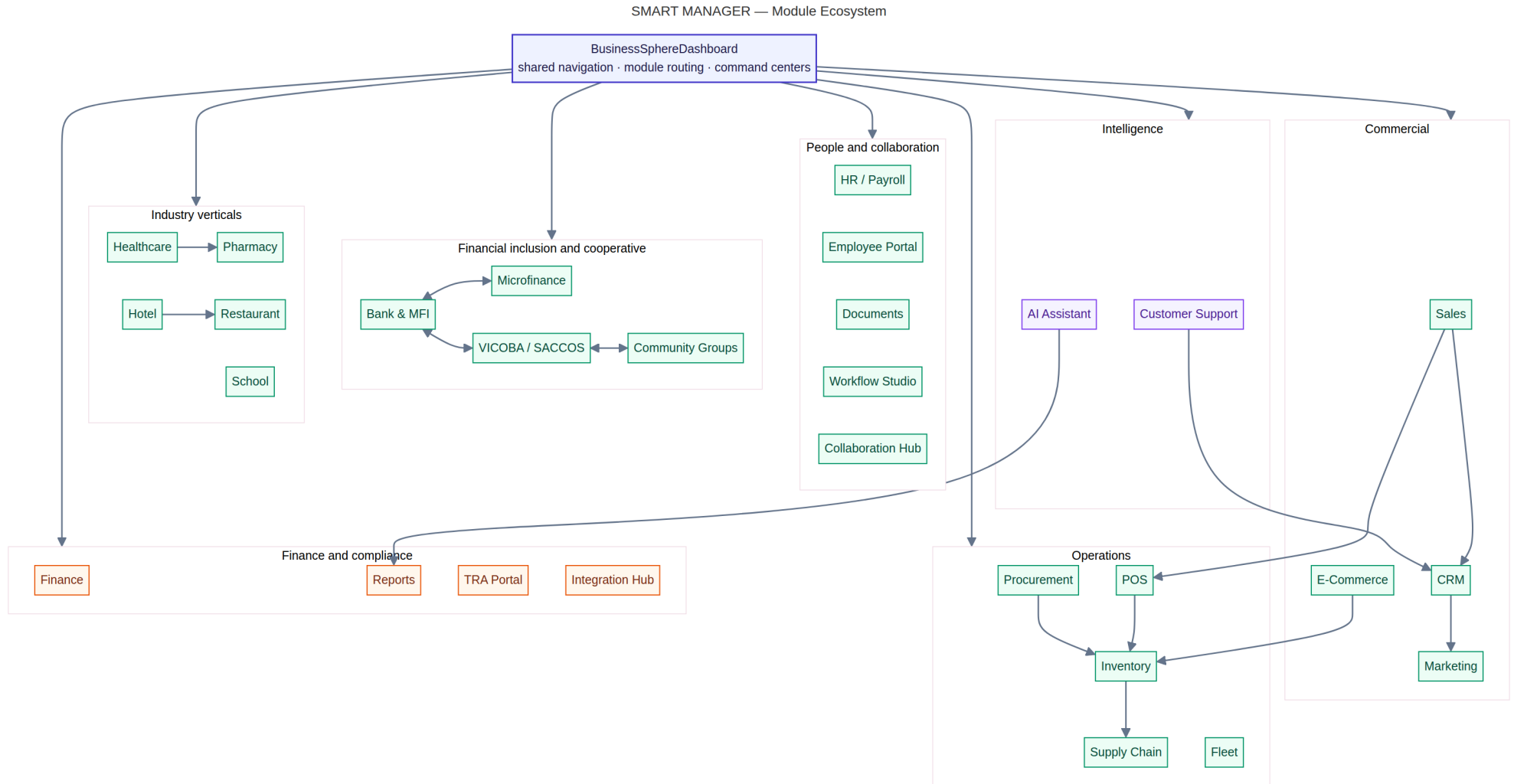
SMART MANAGER — Application Layers



Senior developer interpretation. The layered view shows the request path from React workspaces through typed tRPC contracts, protected Express procedures, domain services, deterministic financial calculations, security checks, persistent Supabase records, storage references, and scheduled callbacks. Each layer has a distinct responsibility and test seam.

3. Module ecosystem

SMART MANAGER ERP · focused architecture view

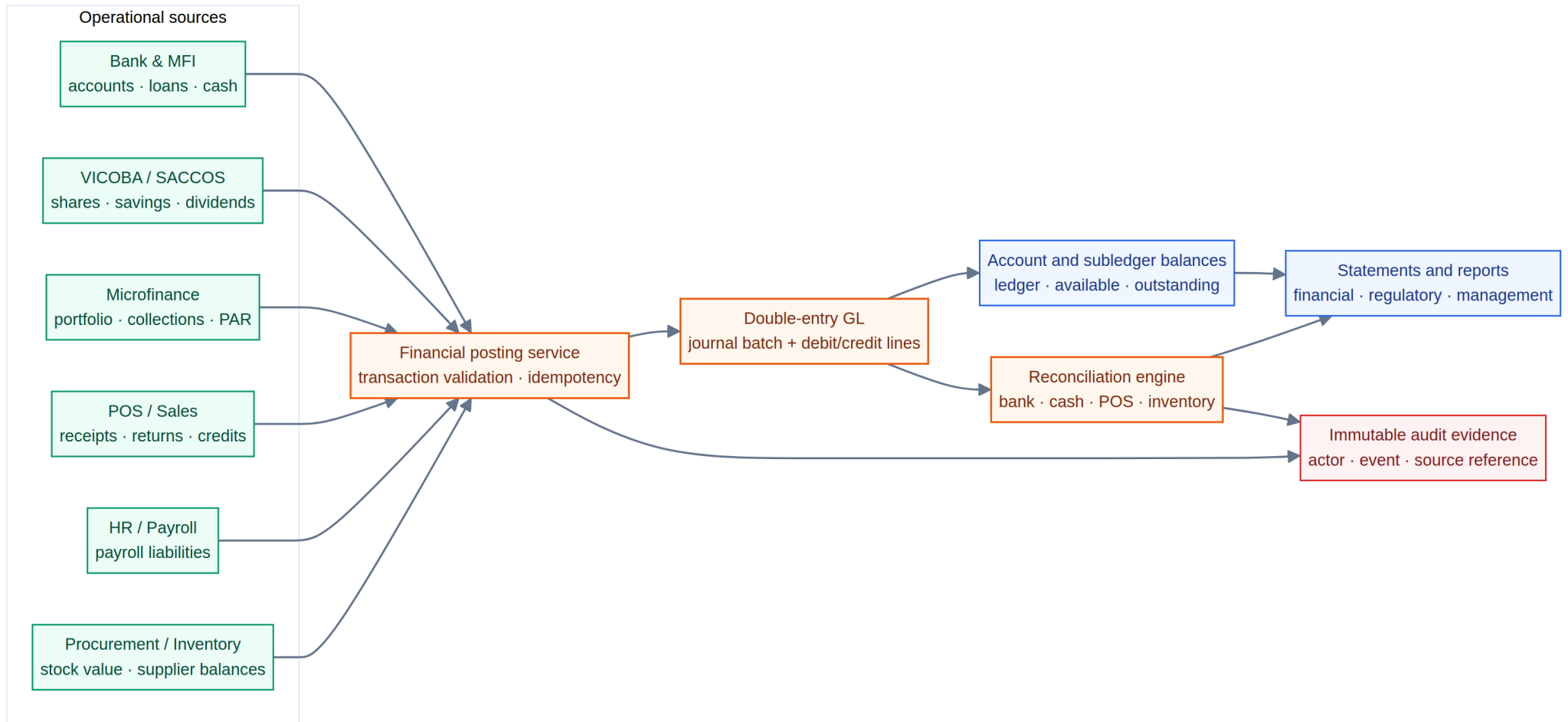


Senior developer interpretation. The ecosystem view shows the ERP as a shared shell with domain groups rather than a collection of isolated screens. Commercial, operations, finance, people, inclusion, vertical, and intelligence modules converge on common navigation, contracts, permissions, data, and reporting while retaining bounded business responsibilities.

4. Shared financial and data plane

SMART MANAGER ERP · focused architecture view

SMART MANAGER — Shared Financial and Data Plane

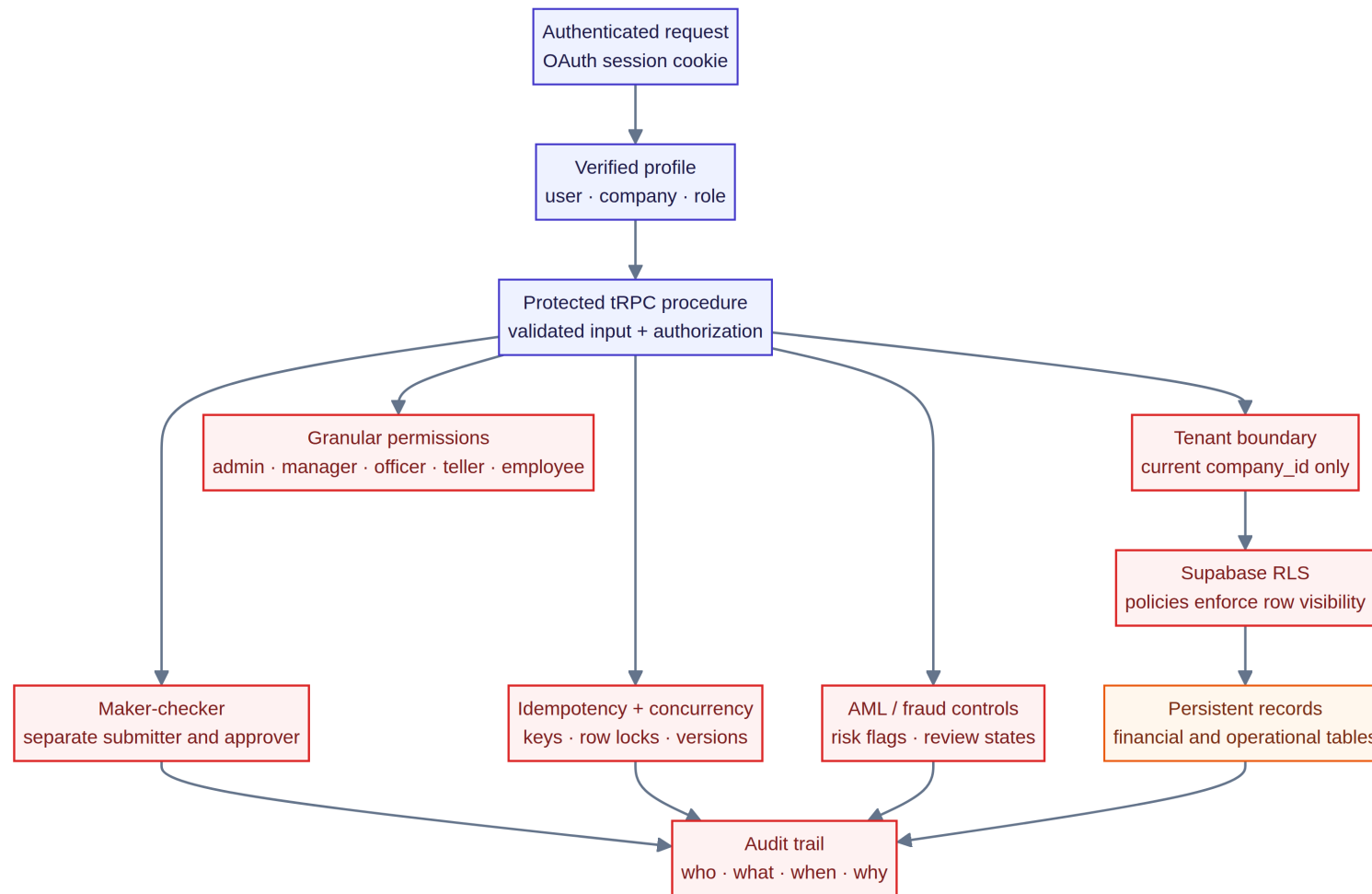


Senior developer interpretation. This is the accounting spine. Operational modules produce validated business events; the posting service converts them into balanced journal entries and subledger changes; reconciliation and reporting consume controlled balances. Audit evidence is emitted alongside mutations, not reconstructed after the fact.

5. Security, tenant isolation, and governance

SMART MANAGER ERP · focused architecture view

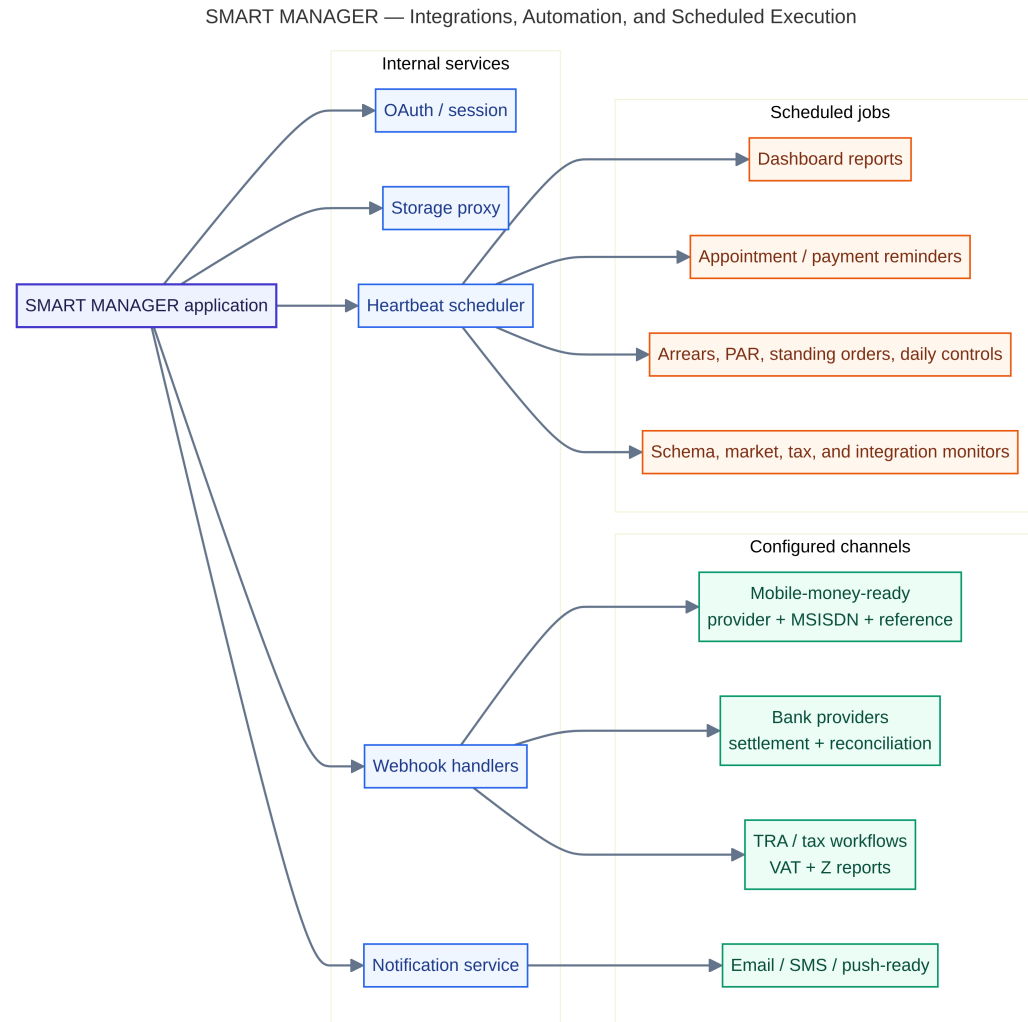
SMART MANAGER — Security, Tenant Isolation, and Governance



Senior developer interpretation. The security view represents defense in depth. A session is converted into a verified profile, then every protected procedure applies input validation, company and role authorization, database RLS, maker-checker separation, idempotency, concurrency controls, AML review states, and audit recording.

6. Integrations, automation, and scheduled execution

SMART MANAGER ERP · focused architecture view

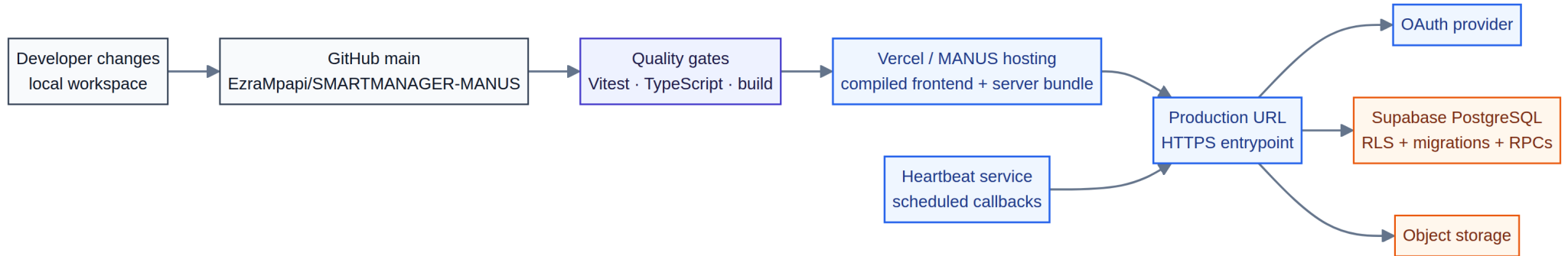


Senior developer interpretation. The integration view distinguishes internal adapters from external channels. Provider workflows remain status-driven and reference-based until settlement is verified. Heartbeat callbacks handle recurring reports, reminders, portfolio controls, standing-order processing, and monitors without relying on an unsafe ad-hoc timer.

7. Deployment and runtime topology

SMART MANAGER ERP · focused architecture view

SMART MANAGER — Deployment and Runtime Topology



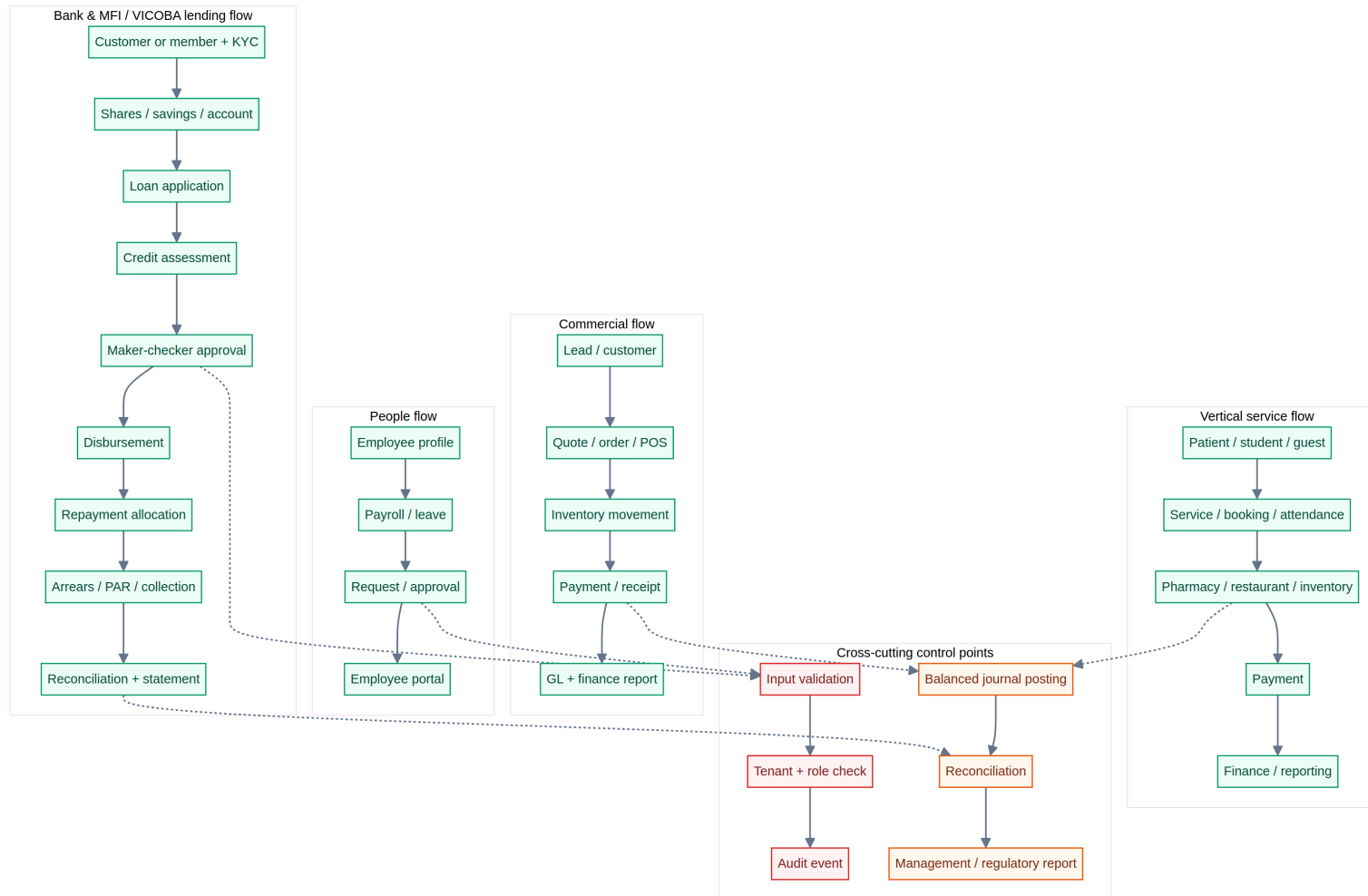
Senior developer interpretation. The runtime view describes the delivery chain from source changes to GitHub, quality gates, hosting, and the production HTTPS endpoint. Runtime dependencies include OAuth, Supabase migrations/RPCs/RLS, object storage, and scheduled callbacks. A deployment is not considered complete until the hosted endpoint is verified.

Figure 7 of 8 · Source: 07_deployment_runtime.mmd

8. Critical end-to-end workflows

SMART MANAGER ERP · focused architecture view

SMART MANAGER — Critical End-to-End Workflows



Senior developer interpretation. The workflow view connects the major business journeys to shared controls. Lending covers KYC through repayment and arrears; commercial operations cover order through stock, payment, and GL; people operations cover employee lifecycle and approvals; vertical services cover service delivery and settlement. Validation, tenant checks, balanced posting, reconciliation, audit, and reporting are cross-cutting gates.